AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

APRIL/ MAY 2017

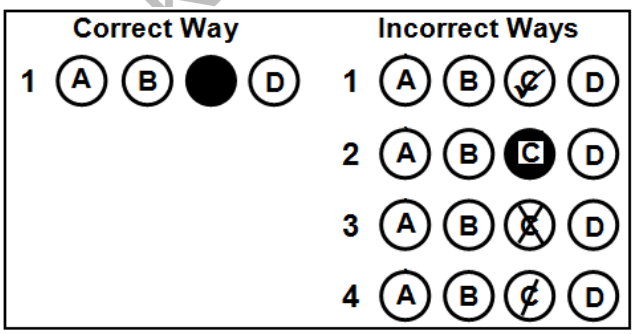
Physics Paper I

Time: 35 minutes Marks: 25

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

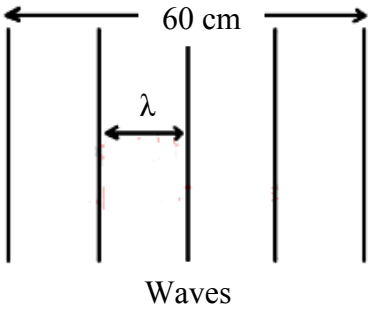
1. Which of the following depicts the appropriate change in wavelength and speed of water waves, when they pass from deep to shallow water?

	Wavelength	Speed
A	Decreases	Decreases
B	No change	Decreases
C	Increases	No change
D	Increases	Increases

2. Which of the following characteristics of musical sound depends upon the frequency?

A. Pitch
B. Quality
C. Loudness
D. Amplitude

3. In the given figure, the frequency of the wave source is 3 Hz. The velocity of the wave will be



A. 0.5 cm/s
B. 12 cm/s
C. 20 cm/s
D. 45 cm/s

4. A sound wave is a mechanical wave. This means that it CANNOT transport energy through

A. solids.
B. liquids.
C. plasma.
D. vacuum.

5. The purpose of using convex lenses in an astronomical telescope is to

A. reflect light rays from distant objects.
B. absorb light rays from distant objects.
C. diverge light rays from distant objects.
D. converge light rays from distant objects.

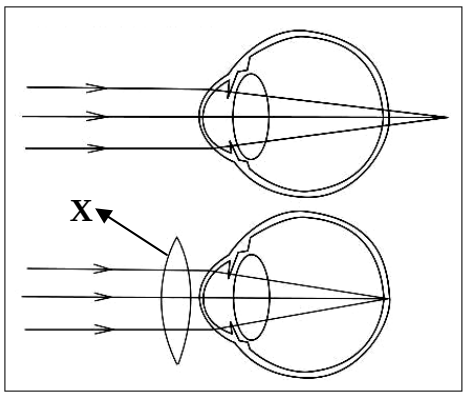
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6. Which of the following defect(s) in vision occurs when the aperture of an eye ball increases?

I. Astigmatism
II. Long-sightedness
III. Short-sightedness

A. I only
B. II only
C. III only
D. I and II

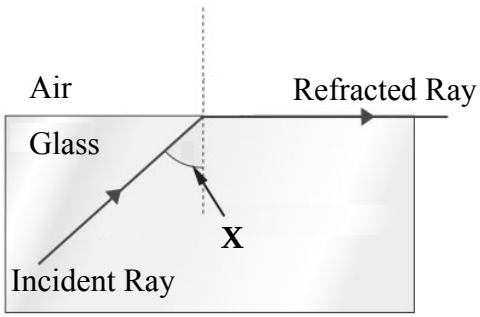
7. In the given ray diagram, a defect and its correction in a human eye is shown.



The optical media **X** used to correct this defect is a

A. convex lens.
B. concave lens.
C. convex mirror.
D. concave mirror.

8. In the given diagram, the angle **X** is called the



A. right angle.
B. critical angle.
C. angle of reflection.
D. angle of deviation.

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9. If two capacitors of 8 μF each are connected in parallel with one another, then their effective capacitance will be

   A. 0.12 μF
   B. 0.25 μF
   C. 8 μF
   D. 16 μF

10. If three resistors of 6 Ω each are connected in series combination, then the value of effective resistance will be equal to

   A. – 6 Ω
   B. 0 Ω
   C. 0.5 Ω
   D. 18 Ω

11. If current of 20 A flows through the live wire, then the current in the neutral wire will be

   A. 0 A
   B. 10 A
   C. 20 A
   D. 40 A

12. If a 200 Watt bulb glows for 1,000 hours, then the energy consumed by it will be

   A. 200 kWh
   B. 800 kWh
   C. 1,200 kWh
   D. 8,00,000 kWh

13. If the speed of a conductor through a magnetic field increases, then the induced electromotive force (e.m.f.) in the conductor

   A. increases.
   B. decreases.
   C. becomes zero.
   D. remains constant.

14. If a conductor is placed in a stationary and non-varying magnetic field, then the induced electromotive force (e.m.f.) will be

   A. low.
   B. high.
   C. zero.
   D. variable.

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15. If the current through a coil of an electromagnet reverses, then the

    A. magnetic field increases.
    B. magnetic field decreases.
    C. direction of the magnetic field reverses.
    D. direction of the magnetic field remains the same.

16. An alternating current (AC) generator is an example of

    A. inductor.
    B. impedance.
    C. mutual induction.
    D. electromagnetic induction.

17. In electronics, the characteristics of an analogue quantity is that it

    A. randomly selects sets of quantities.
    B. can be represented by discrete levels.
    C. has a continuous set of values over a given range.
    D. can be described with a finite number of mathematical steps.

18. Which of the following is/ are the basic operation(s) performed by a computer?

    I. Word processing
    II. Non-arithmetic operations
    III. Monitoring and controlling data

    A. I only
    B. II only
    C. I and III
    D. I, II and III

19. To collect information for a special purpose, store it in a computer and retrieve when needed is called

    A. data managing.
    B. word processing.
    C. communication technology.
    D. monitoring and controlling data.

20. Which of the following is the main use of word processing?

    A. Typing
    B. Monitoring
    C. Controlling
    D. Data managing

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21. The ionising power of gamma rays is

    A. greater than beta particles.
    B. greater than alpha particles.
    C. less than both alpha and beta particles.
    D. less than alpha but greater than beta particles.

22. The given equation shows the disintegration of Ra^{226} into Rn^{222} when a radioactive emission occurs. The radioactive emission is represented by X. What do you infer about X ?

$$^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + {}^{A}_{Z}X$$

    A. A beta particle is emitted.
    B. An alpha particle is emitted.
    C. Only gamma radiation is emitted.
    D. 4 protons and 2 neutrons are emitted from the nucleus.

23. If a nucleus of an element has atomic mass 24, then it will have

    A. 11 protons and 13 neutrons.
    B. 11 protons and 13 electrons.
    C. 11 electrons, 14 protons and 10 neutrons.
    D. 11 electrons, 13 protons and 11 neutrons.

24. A reaction which occurs on the surface of the Sun is

    A. fusion.
    B. fission.
    C. annihilation.
    D. pair production.

25. Large amount of energy in controlled form is released in

    A. fusion reaction.
    B. fission reaction.
    C. physical reaction.
    D. endothermic reaction.

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